

Genre Prediction Using Neural Networks

Predicting movie genres from metadata and descriptions.

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MOVIE

Problem Statement

1

Goal

Predict movie genres.

Use structured and textual features.

2

Why It Matters

Multi-label genres are complex.

Useful for recommendations.

Aids categorization.

3

Challenges

Mixed data types exist.

Multiple labels per movie.



Dataset Overview

Dataset Information

- ~10000 movies analyzed.
- Features: overview, score, budget.
- Additional features: language, country, genre.

Example Movie Row

Overview: "A cyborg is sent from the future..."

Genres: Action, Sci-Fi, Thriller.

Decades of Cinema: Trends in Movie Production & Revenue

Total Movies

9K

TopEarningCountry

AU

Revenue_Per_Movie_By_Year

252.72M

MostFrequentGenre

Drama

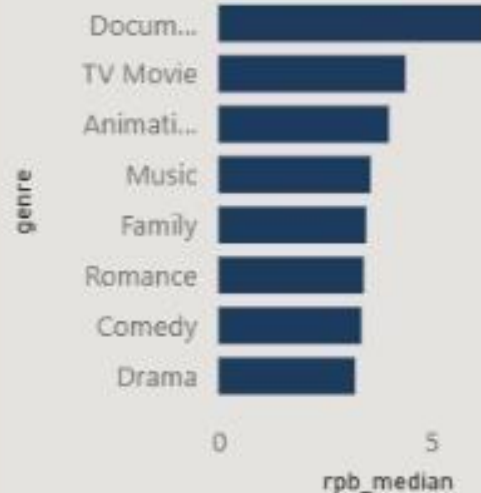
country

All

year

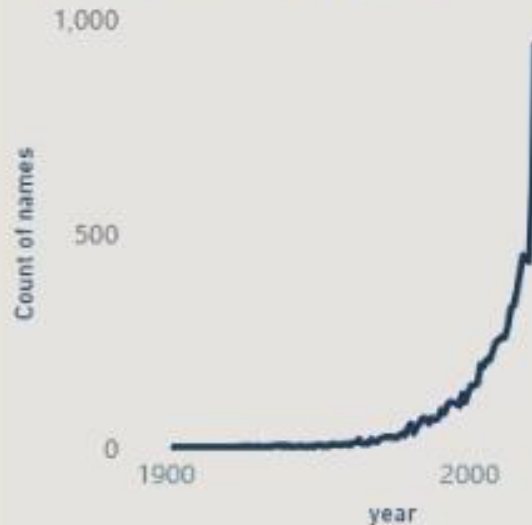
All

rpb_median by genre

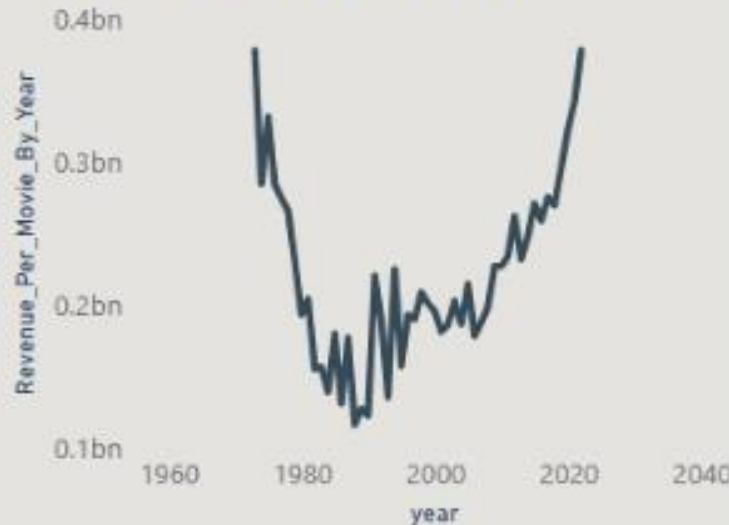


There is a significant and rapid increase in both revenue and the number of movies produced each year.

Count of Movies by year



Revenue to movie count by year



Total Revenue by country



The United States and Australia are the leading countries in terms of film production, consistently generating the highest number of films each year.

Data Preprocessing

This crucial phase refines raw movie data. It ensures quality and consistency. The dataset is now ready for modeling.

Load & Initial Clean

Data loaded, NaNs handled.
Whitespace is stripped.

Filter & Validate

Remove invalid entries. Filter by
revenue, status and year.

Deduplicate & Store

Unique records kept. Cleaned data
saved to database.



Feature Engineering



Text Features

Overview encoded.

Used SentenceTransformer.



Structured Features

Numeric: score, budget,
revenue.

Categorical: language, country.

OneHotEncoder used.



Target Encoding

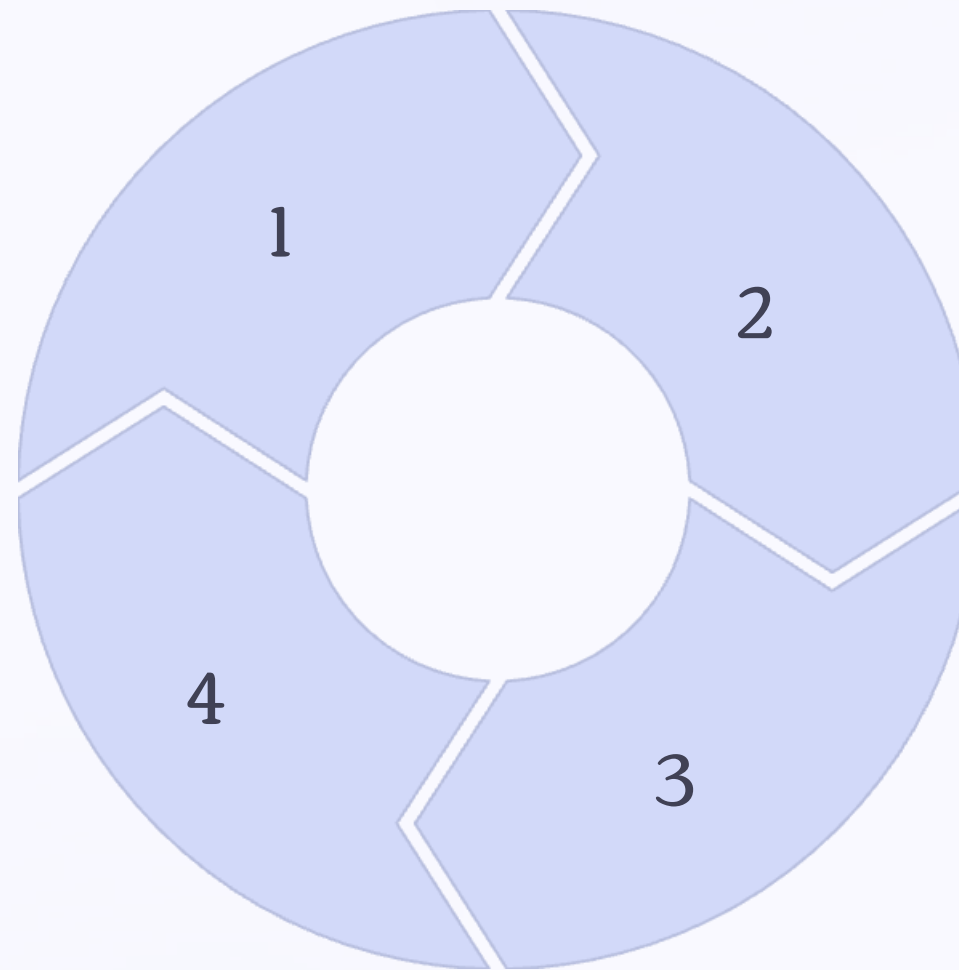
MultiLabelBinarizer applied.

Genre column transformed.

Model Architecture

Model Type
Neural Network.
Keras Sequential.

Loss & Threshold
Binary Crossentropy loss.
Prediction cutoff: 0.35.



Hidden Layers

Dense (384), BatchNorm,
Dropout(0.4).
Dense (192), Dropout(0.3).

Output Layer

Sigmoid activation.
For multi-label prediction.

Model Performance

0.6429

Micro F1-score

Overall prediction accuracy.

0.5712

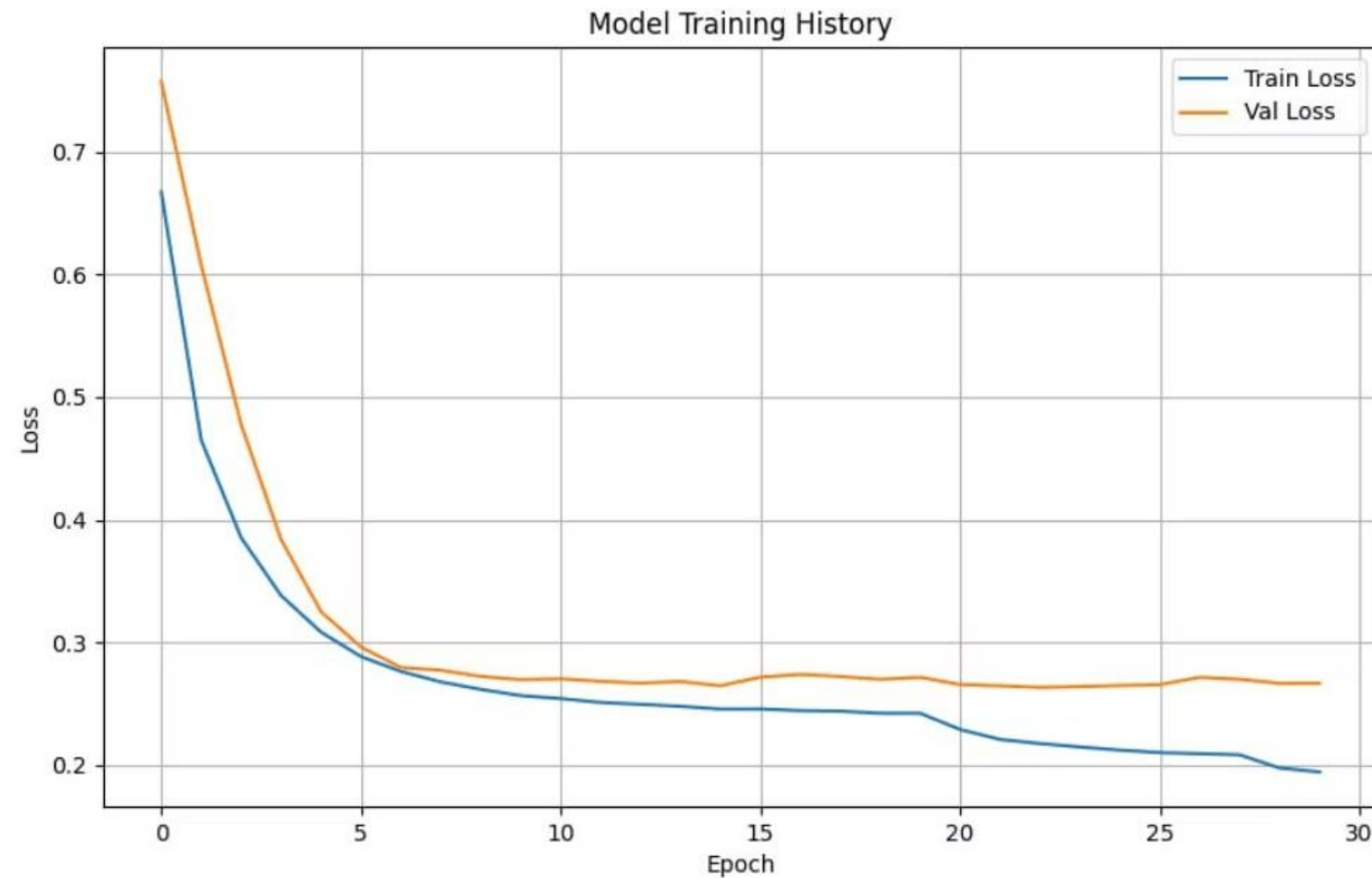
Macro F1-score

Average per-genre accuracy.

0.1016

Hamming Loss

Fraction of incorrect labels.



Summary

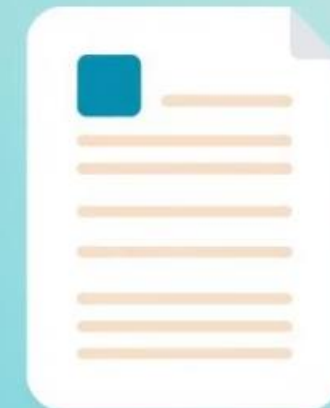


Achievements

Multi-label classifier built.

Used embeddings and features.

MLflow integrated.



A photograph of a presentation screen displaying the text "Thank you" in a blue, sans-serif font. The screen is set against a light blue background. In the foreground, the backs of several audience members are visible, some of whom are clapping. The room is dimly lit, with the primary light source being the screen.

Thank you

🙏 Thank You

Any questions?